

Total No. of Questions : 8]

SEAT No. :

P-6676

[Total No. of Pages : 3

[6181]-244

B.E. (Mechanical Engineering)
COMPUTER INTEGRATED MANUFACTURING
(2019 Pattern) (Semester - VIII) (402048)

Time : 2½ Hours]

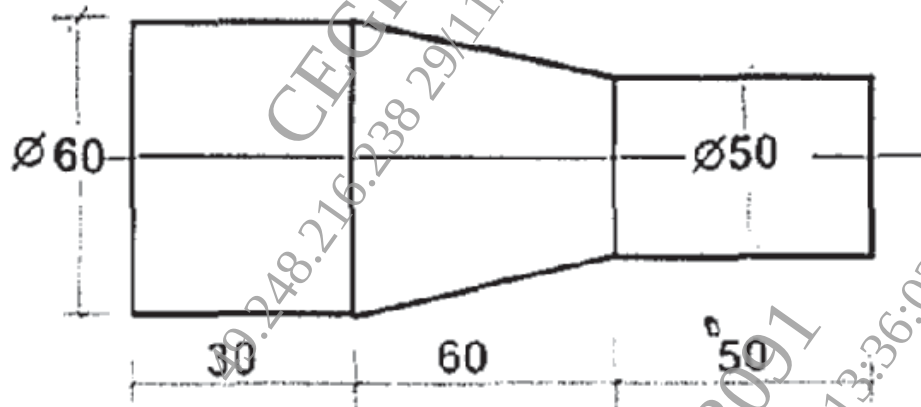
[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume Suitable data, if necessary.

Q1) a) Draw & Explain the components of CNC Machine Tool. [9]

b) Write a CNC turning Program for following. [9]



(All dimensions are in mm)

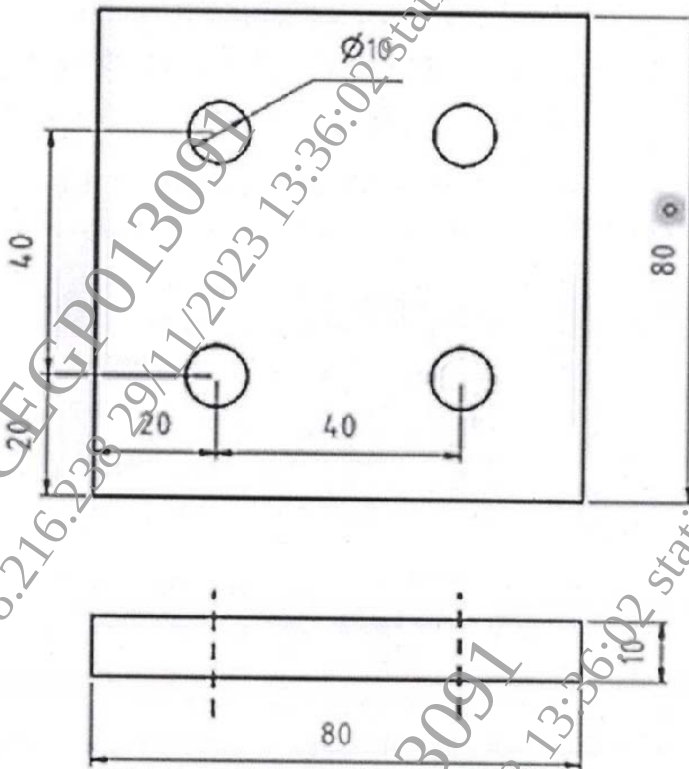
Fig. 1b

OR

Q2) a) Explain Tool length and cutter radius compensation with suitable G-codes. [9]

P.T.O.

- b) Write a part Program for 10 mm drilling operation at four location on milling machine on a work piece shown in Figure 2b. [9]



(All dimensions are in mm)

Fig 2b

- Q3) a)** Describes the different approaches to Computer Aided Process Planning (CAPP). [9]

- b) Explain in brief Capacity planning and Enterprise Resource Planning (ERP). [8]

OR

- Q4) a)** Explain Logical steps in Computer Aided Process Planning (CAPP). [8]

- b) Describe the Manufacturing Resource Planning (MRP-II) with input, working, outputs and benefits. [9]

- Q5) a)** Explain any three Flexible Manufacturing Systems (FMS) based on layout with suitable sketches. [9]

- b) Explain Optiz parts coding system with suitable example. [9]

OR

Q6) a) Describe Rank order Clustering (ROC) algorithm with suitable example. [9]

b) Explain cellular manufacturing and types of machine cell design with layouts. [9]

Q7) a) Explain the classification of components of Industry 4.0. [9]

b) Explain Big-Data and Cloud Computing used for IoT along with its benefits and limitations. [8]

OR

Q8) a) Explain the use of IoT for Smart Manufacturing, Predictive Maintenance and Supply-Chain & Logistics applications. [9]

b) Explain with an example, “How Digital Twin is implemented for Smart Manufacturing?” [8]
